CAPITAL ALLOCATION PRINCIPLES AND PITFALLS

Learning Outcome

- describe principles of capital allocation and common capital allocation pitfalls

Capital Allocation Principles

While the analytical tools and investment decision criteria introduced in the prior lesson are quantitative and straightforward, there is considerable latitude for errors and misjudgment. To improve the decision-making process, key principles should be followed when using these tools.

- *After-tax cash flows:* Managers should evaluate capital allocation decisions based on after-tax cash flows rather than other profit- or accounting-based measures. Managers must reflect the impact of taxation on a project's expected cash flows, specifically the tax benefits derived from non-cash deductions, such as depreciation and amortization.
- *Incremental cash flows only but examine broadly:* Capital allocation analysis should ignore **sunk costs**, or those expenses that have already been incurred (or written off), and include only incremental cash flows associated with a new investment as compared to without it. However, capital investments often have an impact on the rest of the firm, which may be positive or negative. A positive effect might include cost savings with business activities that directly result from making the investment, while a negative effect might be the loss of sales from a similar product. Both are incremental cash flows, so they should be included in the analysis.
- *Timing of cash flows:* The forecasted timing, duration, volatility, and change in the possible direction of the expected cash flows must be considered for a capital investment. Notice how the NPV and IRR can change when cash flows are moved from one period to another.

Capital Allocation Pitfalls

Despite adhering to the principles, capital allocation is challenging for most firms. We divide common capital allocation pitfalls into cognitive errors and behavioral biases. Cognitive errors include calculation and other mistakes, while behavioral biases include errors in judgment and blind spots.

Cognitive Errors in Capital Allocation

- *Internal forecasting errors:* Management may make errors in their forecasts, which may be difficult to impossible for external analysts to identify. However, if significant enough, the incorrect or flawed analysis will ultimately manifest itself in failed, or underperforming, investments. Forecasting errors include incorrect cost or required rates of return inputs. For example, overhead costs such as management time, information technology support, financial systems, can be challenging to estimate. Finally, companies often fail to incorporate competitor responses into the analysis of a planned investment.
- *Ignoring costs of internal financing:* The primary source of financing for investments by large corporate issuers is cash flows from operations (i.e., not borrowing or issuing shares). Many management teams behave as if these internally generated funds are "free" but scarce and allocate them according to a budget that is closely tied to prior-period amounts. External financing from debt or equity issuance, however, is treated differently: It is used less often, typically only for larger investments, such as acquisitions, and treated as "expensive." This is flawed logic. Internally generated capital, such as cash flow from operations, *is* equity financing because it could be

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Regardless of financing source, management should use appropriate risk-adjusted required rates of return to evaluate capital investments; those funded by internally generated funds do not automatically get a lower *r*.

This error is hard to detect and isolate from other errors and biases, but analysts should inquire about management's capital allocation process and how the source of financing influences investment decisions. If a company has an aversion to raising external capital and to returning capital, management may potentially be affected by this.

- *Inconsistent treatment of, or ignoring, inflation:* Inflation affects capital allocation in several ways. The first is whether the investment analysis is done in nominal or in real terms. Nominal cash flows include inflation effects, whereas real cash flows are adjusted downward to remove the effect of inflation (or adjusted upward to remove the effect of deflation). Companies may perform analysis in either nominal or real terms, but the approach to cash flows and discount rate should be consistent. That is, nominal cash flows should be discounted at a nominal discount rate, and real cash flows should be discounted at a real rate.

Investment analysis embeds, explicitly or implicitly, expectations for inflation, which probably does not affect all unit prices and unit costs uniformly. For example, rising oil prices are obviously beneficial for oil producers, which sell their product at a higher price. However, rising oil prices over longer time periods can be associated with rising production and capital costs, eroding some of the benefit. Second, many oil companies also own refining and chemical businesses, for which crude oil is the main input. Profitability of those businesses tends to decline as crude prices rise. If actual inflation differs from expected inflation, after-tax cash flows will be better or worse than expected depending on how specific sales outputs or cost inputs are affected.

Behavioral Biases in Capital Allocation

- *Inertia:* In a study of more than 1,600 US listed companies, researchers from McKinsey found a 0.92 correlation between levels of capital investment in a business segment or unit from one year to the next.[1] This is the result of management anchoring capital investment budgets to prior-year amounts. Analysts identify this bias by examining the level of capital investment in total, by segment, or by business line, if disclosed, and comparing it to the prior year and the return on investment. If capital investment each year is static or rising despite falling returns on investment, analyst should question the issuer's justification for a capital investment and whether management should consider alternative uses.
- *Basing investment decisions on accounting measures, such as EPS:* Managers often have an incentive to increase accounting measures, such as earnings per share, net income, or return on equity. Many capital investments, even those with high NPVs, can reduce rather than increase these accounting results in the near term, while cost cutting and share buybacks, in contrast, may have a positive effect on such measures. Paying too much attention to short-run accounting numbers can lead a company to choose investments that are not in the long-run interests of its shareholders. Analysts may observe this behavior first by examining the direct financial incentives of management based on the structure and composition of their compensation. Second, analysts can compare the level of capital spending to historical and peer levels to judge whether management has prioritized shorter-term, accounting-based measures. That said, lower capital investment may be a sign of limited investment opportunities, in which case allocating capital to alternative uses is a wise decision.
- *Pet project bias:* Projects that receive preferential treatment, or so-called **pet projects**, are sometimes selected without thorough capital allocation analysis. In other cases, such analysis is conducted but overly optimistic projections are used to inflate the pet project's profitability. Identifying pet projects is difficult, because financial statements are usually aggregated and such projects may not meet the threshold of materiality. Instead, analysts should evaluate the corporate governance structure for warning signs that increase the chances of misallocation of capital: controlled companies or

stakeholders' interests.

- *Failure to consider investment alternatives or alternative scenarios:* While investment idea generation is the first step in the capital allocation process, many viable alternatives are never even considered at some companies. Firms also often fail to consider different outcomes, which can and should be incorporated through breakeven, scenario, and simulation analyses. This error may stem from limited capital investment experience, such as not making divestitures or acquisitions, or no experience with a failed investment. While failure is obviously undesirable, the lack of failure over time may reflect a management team that is not taking enough risk.

Question Set

1. True or false: Investment projects funded using internally generated funds (e.g., cash flow from operations) should be evaluated using a lower required rate of return as compared to projects funded using debt or share issuance.

Hide Solution

Solution:

False. Ignoring the cost of internal financing is a common cognitive error. Internally generated capital, such as cash flow from operations, is equity financing because it could be returned to equity investors as a dividend or share repurchase. While internally generated capital is not raised from equity investors by issuing shares, it is withheld from them, therefore incurring their opportunity cost. Regardless of financing source, management should use an appropriate risk-adjusted required rate of return to evaluate capital investments.

2. Alexandra Tolonen, an investment analyst, prepares forecasts for expansion of a car battery plant and wants to consider the effect of rising lithium carbonate prices. Tolonen adjusts future after-tax cashflows downward to remove the effects of an expected lithium carbonate price increase of 5% and uses real discount rates to calculate the project's NPV. In the second year of the project, the actual lithium carbonate price increased by 10%.

True or false. Given that Tolonen has used inflation-adjusted cash flows and discount rates, the rise in lithium carbonate prices will not impact the project's profitability.

Hide Solution

Solution:

False. Even though Tolonen was consistent in using real cash flows and real rates while preparing the forecast, this fact does not make the project immune to changes in inflation. If actual inflation differs from expected inflation, after-tax cash flows will be better or worse than expected, depending on how specific cost inputs are affected.

3. An analyst is analyzing company XYZ and has gathered annual invested capital and ROIC for each of the three XYZ business segments.

	Capital Expenditures ($ m)			ROIC		
Segment	**20X0**	**20X1**	**20X2**	**20X0**	**20X1**	**20X2**
A	264	282	303	7.50%	7.10%	6.80%
B	297	318	340	10.00%	8.90%	7.10%
C	211	226	242	6.90%	7.80%	9.00%

A. Inertia
B. Sunk cost
C. Pet project

Hide Solution

Solution:

A is correct. Inertia can be identified by examining the level of capital investment in total, by segment, or by business line, if disclosed, and comparing it to the prior year and the return on investment. If capital investment each year is static or rising despite falling returns on investment, the analyst should question the issuer's justification for a capital investment and whether management should consider alternative uses. In the case of XYZ, we can observe that for segments A and B, ROIC is decreasing, but both segments are getting a higher capital allocation each year.

B is incorrect because sunk cost is related to the capital allocation process in determining a potential project's profitability. Sunk costs, or those expenses that have already been incurred, should be ignored when evaluating potential projects.

C is incorrect because a detailed view of individual investment projects undertaken by XYZ's management is not provided. Identifying pet projects from outside the firm is difficult, because financial statements are aggregated and such projects may not meet the threshold of materiality.

4. Bradshaw, a financial analyst, prepares a forecast of future expected gross (pre-tax) cash flows for an investment project. Bradshaw also forecasts future depreciation related to that project and assumes a required rate of return 6%. Based on the information provided and a tax rate of 18%, calculate the NPV of the project. In this jurisdiction, depreciation is not deductible for taxes.

Cash Flow and Depreciation Forecast				
Time	**0**	**1**	**2**	**3**
Gross cash flow	–7.50	4.50	4.50	6.00
Depreciation	0.00	–1.00	–1.00	–1.00

Hide Solution

Solution:

To calculate NPV for this project, Bradshaw should first calculate after-tax cash flows and consider depreciation only to the extent that it is tax deductible and reduces taxes, because it is non-cash. After-tax cash flows can be calculated by applying the following formula to all cash flows to be received in the future.

$$\text{After-tax cash flow}_t = \text{Gross cash flow}_t \times (1 - \text{Tax rate}).$$

After-Tax Cash Flow Forecast				
Time	**0**	**1**	**2**	**3**
After-tax cash flow	–7.50	3.69	3.69	4.92

The calculation of NPV is follows:

$$\text{NPV} = -7.5 + \frac{3.69}{(1+0.06)^1} + \frac{3.69}{(1+0.06)^2} + \frac{4.92}{(1+0.06)^3}.$$

$$\text{NPV} = 3.396$$

5. Explain why some managers might reject projects that significantly increase shareholder value (i.e., have a high NPV).

Managers may have an incentive to increase accounting profitability measures, such as earnings per share, net income, or return on equity. Many capital investments, even those with high NPVs, can reduce rather than increase these measures in the near term, while cost cutting and share buybacks, in contrast, may have a positive effect on such measures. If this incentive is strong enough, management may forgo high-NPV projects in favor of actions that increase near-term earnings per share.

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Discussion

What do you want to discuss... (required)

CS

In Q4, the calculation seems off and I can't figure out how they got their after tax cash flows. I would think year 1 would be (4.5) - ((4.5-1)*.18) = 3.87

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MS

In Q4, why have they not considered the impact of depreciation? the explanation mentions the impact of non cash deductibles to be taken

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